FIT3080 Artificial Intelligence
Solutions to Practice Questions for Final Exam – Machine Learning

1. **[Supervised Machine Learning]** Consider the following dataset consisting of 3 binary features and a binary class variable.

Feature 1	Feature 2	Feature 3	Class
1	0	0	0
1	0	1	1
1	1	1	1
1	1	1	1
0	1	1	0
0	0	1	0

   (a) Based on the dataset above, estimate the prior probability of Class 1.
   SOLUTION: $\Pr(\text{Class} = 1) = 3/6 = 0.5$

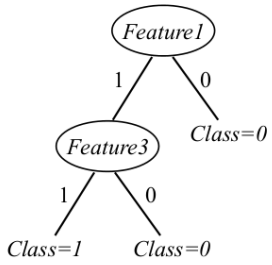
   (b) What is the initial entropy of the class labels over all the data? Show the formula before plugging-in numbers.
   SOLUTION:
   $H(\text{Class}) = -\Pr(\text{Class=1}) \log_2 \Pr(\text{Class=1}) - \Pr(\text{Class=0}) \log_2 \Pr(\text{Class=0})$
   $H(\text{Class}) = -0.5 \log(0.5) - 0.5 \log(0.5) = 1$

   (c) If we were to split the data into 2 groups based on the value of Feature 1, what would be the entropy for each group?
   SOLUTION:
   $H(\text{Class}|\text{Feature1=1}) = -(3/4) \log(3/4) - (1/4) \log(1/4) = .81$
   $H(\text{Class}|\text{Feature1=0}) = -(1) \log(1) - (0) \log(0) = 0$

   (d) What is the Information Gain of Feature 1?
   SOLUTION:

$$
\begin{aligned}
& IG(\text{Feature1}) \\
&= H(\text{Class}) - (\Pr(F1 = 1) H(\text{Class}|F1 = 1) + \Pr(F1 = 0) H(\text{Class}|F1 = 0)) \\
&= 1 - ((4/6) \times 0.81 + (2/6) \times 0) = (1 - 0.54) = 0.46
\end{aligned}
$$

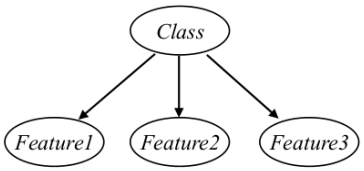
   (e) You are told the Information Gain for Feature 2 and Feature 3 **after splitting on Feature1** is 0.08 and 0.19 respectively. Draw a decision tree for this problem.
   SOLUTION:



   Our second split is on Feature 3, since it has a higher IG than Feature 2. After splitting on Feature 3, we get clean leaf nodes.

(f) We will now build a Naïve Bayes model for predicting the class from these data. Draw the Bayesian Network that corresponds to the Naïve Bayes model.

SOLUTION:



(g) From the data table, write down the conditional probability estimates required for the Naïve Bayes model.

SOLUTION:

$\Pr(\text{Feature1 }=1|\text{Class}=1) = 3/3 = 1$
$\Pr(\text{Feature1 }=1|\text{Class}=0) = 1/3$
$\Pr(\text{Feature2 }=1|\text{Class}=1) = 2/3$
$\Pr(\text{Feature2 }=1|\text{Class}=0) = 1/3$
$\Pr(\text{Feature3 }=1|\text{Class}=1) = 3/3 = 1$
$\Pr(\text{Feature3 }=1|\text{Class}=0) = 2/3$

(h) What class does the Naïve Bayes model predict for a new data item with values (1, 1, 1) for Features 1, 2 and 3 respectively? What is the probability of each class? How does the Naïve Bayes classification compare to the decision tree classification?

SOLUTION:

$$\begin{aligned}
&\Pr(C = 1|F1 = 1, F2 = 1, F3 = 1) \\
&= \alpha \Pr(F1 = 1|C = 1)\Pr(F2 = 1|C = 1)\Pr(F3 = 1|C = 1)\Pr(C = 1) \\
&= \alpha \times 1 \times 2/3 \times 1 \times 1/2 = 1/3 \times \alpha
\end{aligned}$$

$$\begin{aligned}
&\Pr(C = 0|F1 = 1, F2 = 1, F3 = 1) \\
&= \alpha \Pr(F1 = 1|C = 0)\Pr(F2 = 1|C = 0)\Pr(F3 = 1|C = 0)\Pr(C = 0) \\
&= \alpha \times 1/3 \times 1/3 \times 2/3 \times 1/2 = 1/27 \times \alpha
\end{aligned}$$

$\alpha = \frac{1}{1/3+1/27} = 2.7$
$\Pr(C = 1|F1 = 1, F2 = 1, F3 = 1) = 2.7 \times 1/3 = 0.9$
$\Pr(C = 0|F1 = 1, F2 = 1, F3 = 1) = 2.7 \times 1/27 = 0.1$

So, Naïve Bayes assigns Class 1 to case (1,1,1), which is the same as the decision tree, but the probability for Naïve Bayes is 0.9.

2. [**Decision Tree**] Assume that you are given the set of labeled *training examples* below, where each of three features has three possible values: a, b or c. You choose to learn a decision tree from these data.

	F1	F2	F3	Output
ex1	c	b	b	+
ex2	a	a	c	+
ex3	b	c	c	+
ex4	b	c	a	−
ex5	a	b	c	−
ex6	c	a	b	−

What score would the *information gain* calculation assign to feature F1, when deciding which feature to use as the root node for a decision tree being built?

SOLUTION:
$IG(S, F1) = H(S) - \frac{2}{6}H(S_{F1=a}) - \frac{2}{6}H(S_{F1=b}) - \frac{2}{6}H(S_{F1=c})$
where $H(S) = -\frac{3}{6}\log_2\frac{3}{6} - \frac{3}{6}\log_2\frac{3}{6} = 1$
$H(S_{F1=a}) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$
$H(S_{F1=b}) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$
$H(S_{F1=c}) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$
$IG(S, F1) = 1 - \frac{2}{6} \times 1 - \frac{2}{6} \times 1 - \frac{2}{6} \times 1 = 0$

3. [**Naïve Bayes**] Consider a traffic-monitoring agent trying decide whether the stoplights at an intersection are working or not ($W = w$ or $\neg w$). The agent observes two variables, the East-West light's color EW, which is either red or green, and the North-South light's color NS, which is also either red or green. When the lights are functioning, exactly one of the lights is green, while when the lights are broken, both are red (and flashing, but the agent cannot perceive this distinction). The agent has several observations as training data:

NS	EW	W
r	g	w
g	r	w
r	r	$\neg w$
g	r	w
r	g	w
r	g	w
g	r	w

(a) Assume the agent uses a Naïve Bayes classifier to make its prediction. What are the parameters of the model?
SOLUTION:
$Pr(W), Pr(NS|W), Pr(EW|W)$

(b) What is the Maximum Likelihood Estimator (MLE) for the parameters?
SOLUTION:
$Pr(W = w) = 6/7$, $Pr(W = \neg w) = 1 - 6/7 = 1/7$
$Pr(NS = r|W = w) = 3/6 = 1/2$, $Pr(NS = g|W = w) = 1 - 1/2 = 1/2$
$Pr(NS = r|W = \neg w) = 1$, $Pr(NS = g|W = \neg w) = 1 - 1 = 0$
$Pr(EW = r|W = w) = 3/6 = 1/2$, $Pr(NS = g|W = w) = 1 - 1/2 = 1/2$
$Pr(EW = r|W = \neg w) = 1$, $Pr(EW = g|W = \neg w) = 1 - 1 = 0$

(c) Assume the agent observes that both lights are red. What is the probability that the lights are working according to the agent's model?
SOLUTION:
$$\begin{aligned} Pr(W = w|NS = r, EW = r) \\ &= \alpha \times Pr(NS = r|W = w) \times Pr(EW = r|W = w) \times Pr(W = w) \\ &= \alpha \times 1/2 \times 1/2 \times 6/7 = \alpha \times 3/14 \end{aligned}$$
$$\begin{aligned} Pr(W = \neg w|NS = r, EW = r) \\ &= \alpha \times Pr(NS = r|W = \neg w) \times Pr(EW = r|W = \neg w) \times Pr(W = \neg w) \\ &= \alpha \times 1 \times 1 \times 1/7 = \alpha \times 1/7 \end{aligned}$$
$\therefore \alpha = 14/5, Pr(W = w|NS = r, EW = r) = 3/5 = 0.6$

4. [**Perceptron**] Our aim is to train a perceptron with three features f_1, f_2, f_3 and three classes x, y, z. Suppose at some stage in the training, the weights for different classes are as follows:

$$\begin{aligned} w_x &= (-1, -1, -1) \\ w_y &= (1, 1, 1) \\ w_z &= (1, -1, 1) \end{aligned}$$

Write down the updated weights after seeing a new training instance (1,1,1) with label z.

SOLUTION:
$\text{Score}(x) = w_x \cdot f = w_x \cdot (1,1,1) = (-1,-1,-1) \cdot (1,1,1) = -3$
$\text{Score}(y) = w_y \cdot f = w_y \cdot (1,1,1) = (1,1,1) \cdot (1,1,1) = 3$
$\text{Score}(z) = w_z \cdot f = w_z \cdot (1,1,1) = (1,-1,1) \cdot (1,1,1) = 1$

So, the predicted label is y which differs from the true label z, hence we need to update w_z and w_y:
$w_y \leftarrow w_y - f = w_y - (1,1,1) = (1,1,1) - (1,1,1) = (0,0,0)$
$w_z \leftarrow w_z + f = w_z + (1,1,1) = (1,-1,1) + (1,1,1) = (2,0,2)$

5. [**K-nearest-neighbour**] Assume we have the age, loan amount and pay status of some customer from the bank. We need to predict the pay result of a new loan applicant given only age and loan amount. The data are shown below:

Age	Loan Amount (K)	Pay Status	Distance to new customer
20	20	Paid	$\sqrt{(20-42)^2 + (20-140)^2} = 122$
25	45	Paid	$\sqrt{(25-42)^2 + (45-140)^2} = 96.51$
22	95	Default	$\sqrt{(22-42)^2 + (95-140)^2} = 49.24$
35	53	Paid	$\sqrt{(35-42)^2 + (53-140)^2} = 87.28$
35	**120**	**Paid**	$\mathbf{\sqrt{(35-42)^2 + (120-140)^2} = 21.19}$
33	**150**	**Default**	$\mathbf{\sqrt{(33-42)^2 + (150-140)^2} = 13.45}$
40	54	Default	$\sqrt{(40-42)^2 + (54-140)^2} = 86.02$
45	75	Paid	$\sqrt{(45-42)^2 + (75-140)^2} = 65.07$
52	20	Paid	$\sqrt{(52-42)^2 + (20-140)^2} = 120.42$
49	220	Default	$\sqrt{(49-42)^2 + (220-140)^2} = 80.31$
60	**100**	**Default**	$\mathbf{\sqrt{(60-42)^2 + (100-140)^2} = 43.86}$

A new bank customer with age 42 comes in and applies for a loan AUD$140K, how would the 3-nearest-neighbour algorithm predict the pay status of this customer? Paid or Default?

SOLUTION: According to the 3 nearest neighbours (bold-fonted in the table), the algorithm will predict that this new customer will Default on the loan.

6. [**K-means clustering**] Consider the k-means clustering algorithm when answering the following questions.

   (a) What are the two steps performed at each iteration of the clustering algorithm?
   SOLUTION:
   Step 1. Reassign data points to their closest cluster (centre).
   Step 2. Recompute the centre (mean) of each cluster.

   (b) Is the algorithm guaranteed to converge (stop) at some point? And if so, does it converge to a global optimum or a local optimum?
   SOLUTION:
   Yes, convergence is guaranteed, but to a *local* optimum. Other techniques are required to achieve global optimum.

   (c) Choosing the "right number" of clusters for k-means clustering can be difficult. One strategy would be to choose the number of clusters that minimises the sum of the (squared) distances between data points and their cluster centres. Would this be a good strategy? Why?
   SOLUTION:
   No, it is not a good strategy, since the "optimal" number of clusters (i.e., the one that minimizes the sum of the distances to the cluster means) would always be n, the number of datapoints.

   (d) Before running a clustering algorithm it is important to preprocess your data. What preprocessing is most important and why?
   SOLUTION:
   Must rescale your features/dimensions to have the same range, otherwise the distance measure will be controlled by the dimension with the largest scale.

7. Assume we have following data with two attributes, apply the K-means clustering algorithm where K=2 and initialized cluster centroid C1=(1.25, 1.25) and C2=(1.75,4.75). You should use Euclidean distance to perform your calculations.

Instance	D1	D2
X1	1.0	1.5
X2	1.0	4.5
X3	2.0	1.5
X4	2.0	3.5
X5	3.0	2.5
X6	5.0	6.0
X7	4.0	5.5
X8	6.0	1.5

SOLUTION:

	C1	1.25	1.25		
	C2	1.75	4.75		
Iteration 1					
Instance	D1	D2	Distance to C1	Distance to C2	Cluster
X1	1	1.5	0.353553391	3.335416016	C1
X2	1	4.5	3.259601203	0.790569415	C2
X3	2	1.5	0.790569415	3.259601203	C1
X4	2	3.5	2.371708245	1.274754878	C2
X5	3	2.5	2.150581317	2.573907535	C1
X6	5	6	6.051859218	3.482097069	C2
X7	4	5.5	5.062114183	2.371708245	C2
X8	6	1.5	4.756574398	5.35023364	C1
Clusters={X1, X3, X5, X8}, {X2, X4, X6, X7}					
Update centroid	C1	3	1.75		
	C2	3	4.875		
Iteration 2					
Instance	D1	D2	Distance to C1	Distance to C2	Cluster
X1	1	1.5	2.015564437	3.923088707	C1
X2	1	4.5	3.400367627	2.034852575	C2
X3	2	1.5	1.030776406	3.52003196	C1
X4	2	3.5	2.015564437	1.700183814	C2
X5	3	2.5	0.75	2.375	C1
X6	5	6	4.697073557	2.294694969	C2
X7	4	5.5	3.881043674	1.179247642	C2
X8	6	1.5	3.010398645	4.515597967	C1
Clusters={X1, X3, X5, X8}, {X2, X4, X6, X7}					
Clusters are the same for iteration 1 and 2, STOP					